

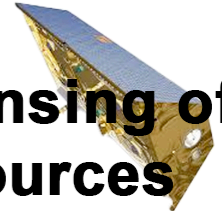


Australian
National
University

Research School
of Earth Sciences



**Remote Sensing of
Water Resources**
EMSC3025/6025



Unsaturated Flow and Miscellaneous Groundwater Topics

EMSC3025/6025

[Back to water-course](#)

EMSC3025/6025: Remote Sensing of Water
Resources

Dr. Sia Ghelichkhan

Objectives

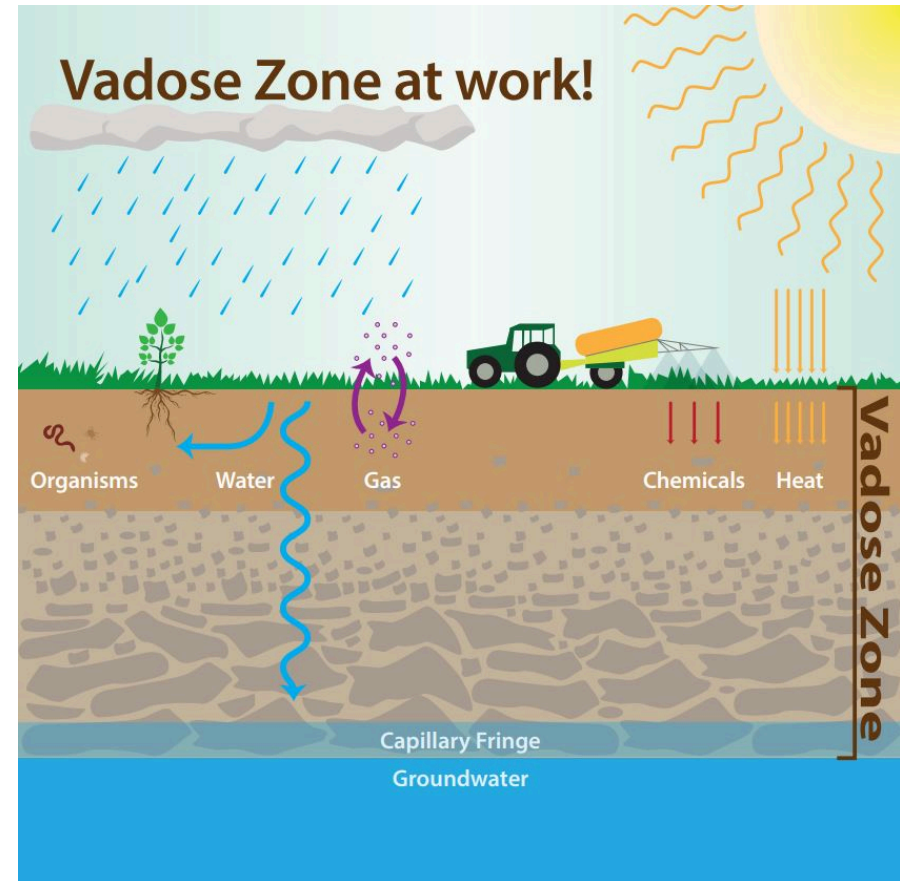
By the end of this lecture, you should be able to:

- Define **volumetric water content** θ and **water saturation** s in the vadose zone.
- Explain why **pressure head** ψ is **negative** in the unsaturated zone.
- Describe methods for **measuring water content** (gravimetric and indirect techniques).
- Understand **infiltration, redistribution, and recharge** processes in unsaturated flow.
- Explain **characteristic curves**: water retention $\theta(\psi)$ and hydraulic conductivity $K(\psi)$.
- Recognize key **soil water models** (Brooks-Corey, van Genuchten) and their applications.
- Apply the **cubic law** for flow in fractured media and calculate fracture hydraulic conductivity.

Unsaturated or vadose zone

The space between the water table and the ground surface is called the vadose zone.

- The flow of water in the unsaturated zone is complicated.
- The the pore spaces contain both soil gas and water.
- Hydraulic constants, such as **conductivity** and **diffusion**, become hydraulic functions.
- The transmissive/storage properties can change dramatically depending on the water content.



Vadose zone essentials

In the **unsaturated (vadose) zone**, pores contain **both soil gas and water**. Two core descriptors:

1. Volumetric water content:

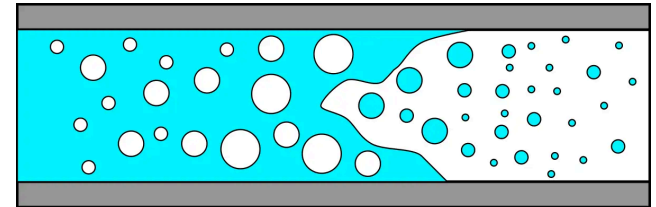
$$\theta = \frac{V_w}{V_T} \quad (0 < \theta < n)$$

V_w : volume of water, V_T : total sample volume, n : porosity.

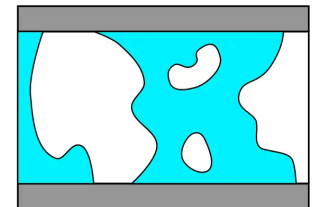
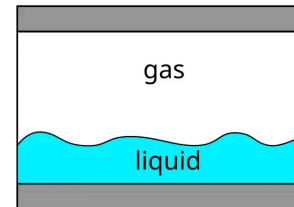
2. Water saturation:

$$s = \frac{V_w}{V_{\text{void}}} \quad (0 < s < 1)$$

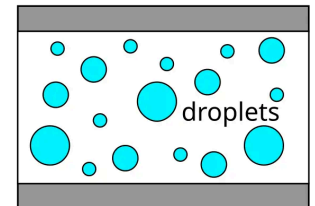
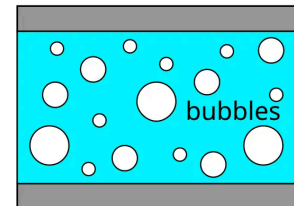
Key point In the unsaturated zone, typically $\theta < n$ and $s < 1$.



a) Transient two-phase flow.



b) Separated two-phase flow.



c) Dispersed two-phase flow.

Measuring water content

- For a direct measurement of water content

θ_w as a weight fraction:

1. Weight the soil in a covered container.
2. Dry it in a convection oven for 24 hours.
3. Weight it again.

$$\theta_w = \frac{W_{wet-soil}}{W_{dry-soil}} - 1$$

Volumetric water content: $\theta = \theta_w \frac{\rho_b}{\rho_w}$

Indirect methods: measuring properties associated with water content.

- Electrical conductivity: Measuring the electrical resistivity of the soil: dependent on water content.
- Time-domain reflectometry: Measuring dielectric constant of the soil: dependent on water content.
- gamma-ray attenuation: Water absorbs more gamma rays than soil.

Hydraulic head in the unsaturated zone

Hydraulic head: determines the direction of fluid flow.

It combines elevation and pressure:

$$h = z + \psi$$

- Above the water table: $\psi < 0$ (**suction/tension**). Water pressure less than atmospheric pressure. Explains why water in unsaturated zone **does not flow to a borehole**.
- At the water table: $\psi = 0$.
- Below the water table (saturated): $\psi > 0$.

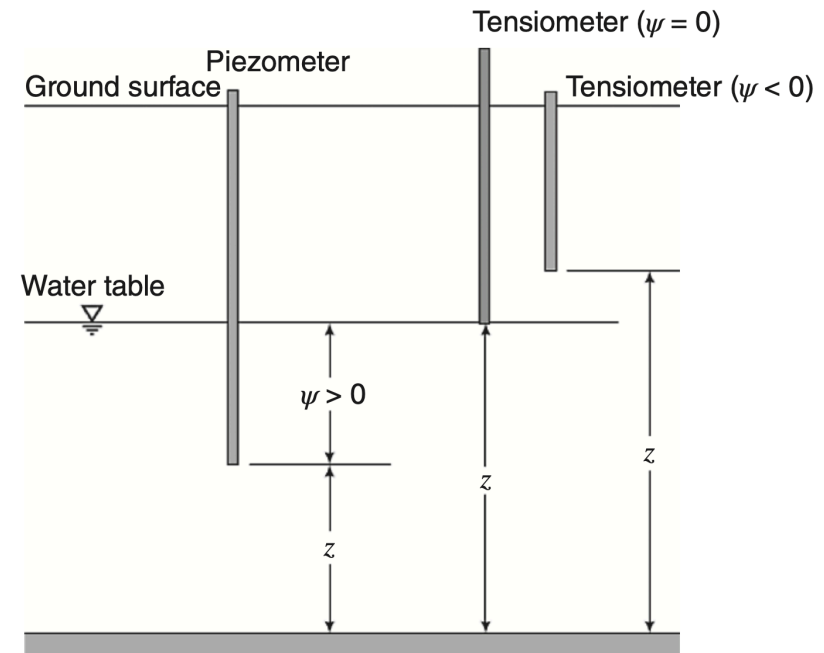


Figure 1: ψ negative above, zero at, and positive below the water table

Measuring negative pressure head — Tensiometer

Tensiometer (ASTM 1992; Stannard 1986):

- Porous **ceramic cup** connected to a water column/manometer, **vacuum gauge**, or **pressure transducer**.
- Cup is water-filled → provides **hydraulic connection** to pore water.
- As soil water pressure changes, instrument records ψ (**negative**) to maintain equilibrium.

Interpreting head profiles

From the text example:

- **Pressure head ψ** : negative **above** the water table, positive **below**.
- **Total head $h = z + \psi$** : decreases with depth through the unsaturated zone $\rightarrow$ **downward flow**.

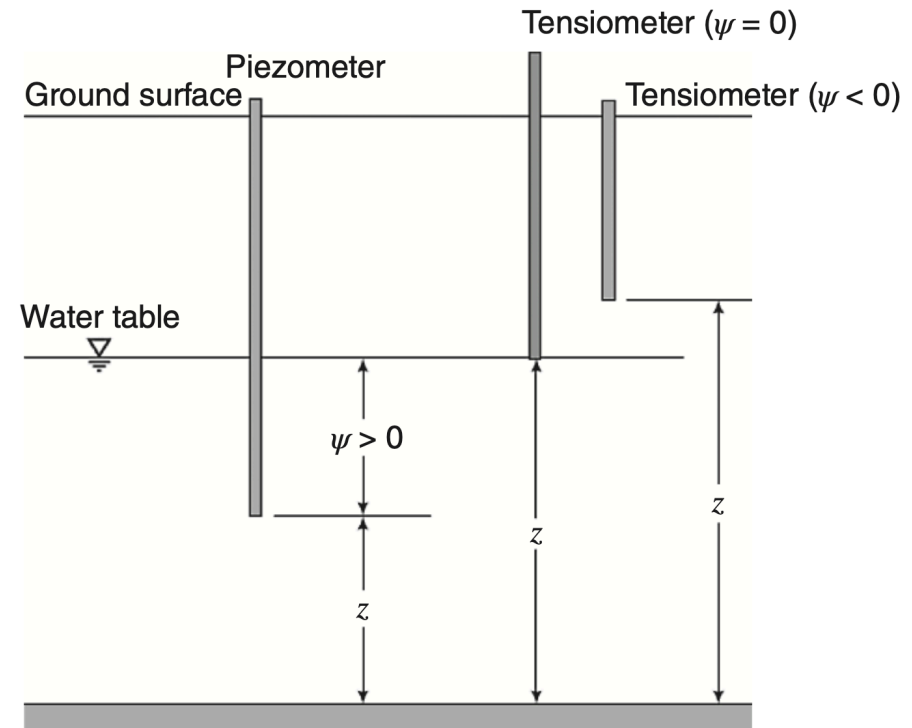


Figure 2: ψ negative above, zero at, and positive below the water table

The process of infiltration is complex

Consider the process of infiltration
precipitation in unsaturated soil!

Changes in Moisture Content During Infiltration

- Water content in the **unsaturated zone** varies with **space** and **time**.
- At early times → **capillary forces dominate**, increasing water content near the surface.
- Later → infiltration ceases, and water redistributes downward.
- Both **capillary** and **gravitational** forces act.
- A **wetting front** forms: narrow zone where water content sharply increases.

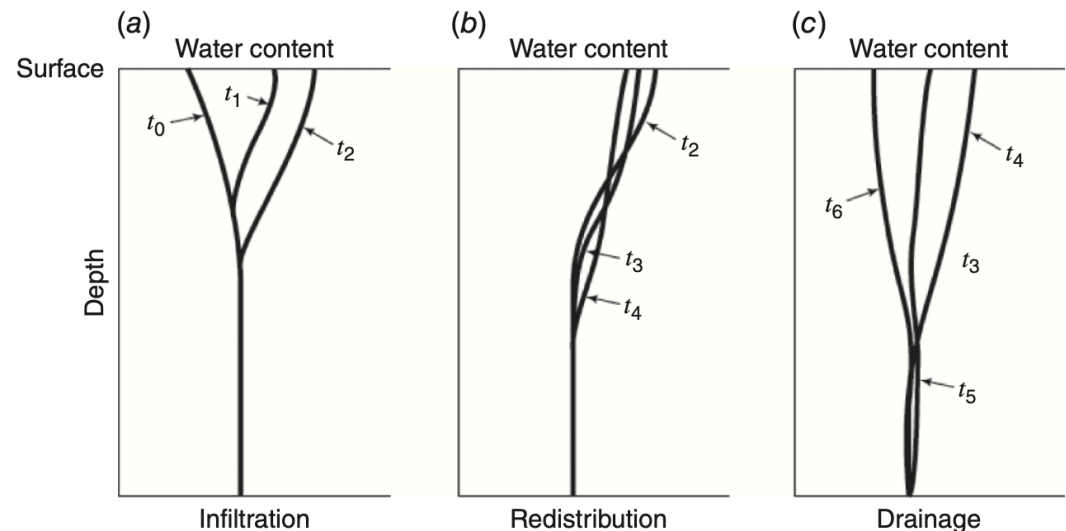


Figure 3: Processes of infiltration redistribution and recharge

Redistribution After Infiltration

- After infiltration stops, the infiltrated pulse moves downward.
- The **wetting front** is clearly visible.
- Soil above the front gradually dries, while water below remains at higher content.
- This redistribution continues until equilibrium is reached or evapotranspiration intervenes.

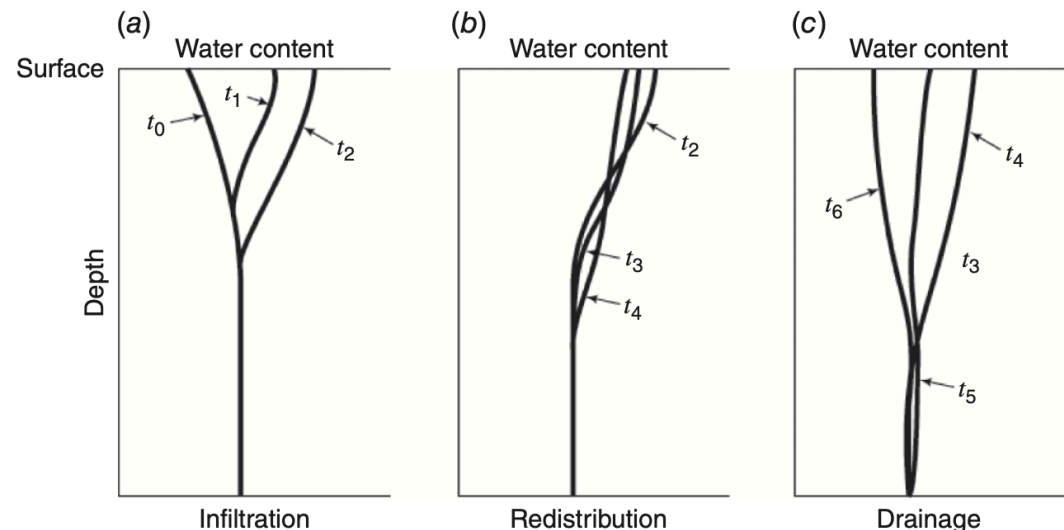


Figure 4: Processes of infiltration redistribution and recharge

Final Stage of Moisture Movement

- Near the surface, **evapotranspiration** removes water once infiltration ends.
- Eventually, infiltrated water reaches the **saturated zone**.
- The unsaturated zone returns toward its initial condition, with soil moisture gradually reduced.
- Figure illustrates soil water entering saturated regions.

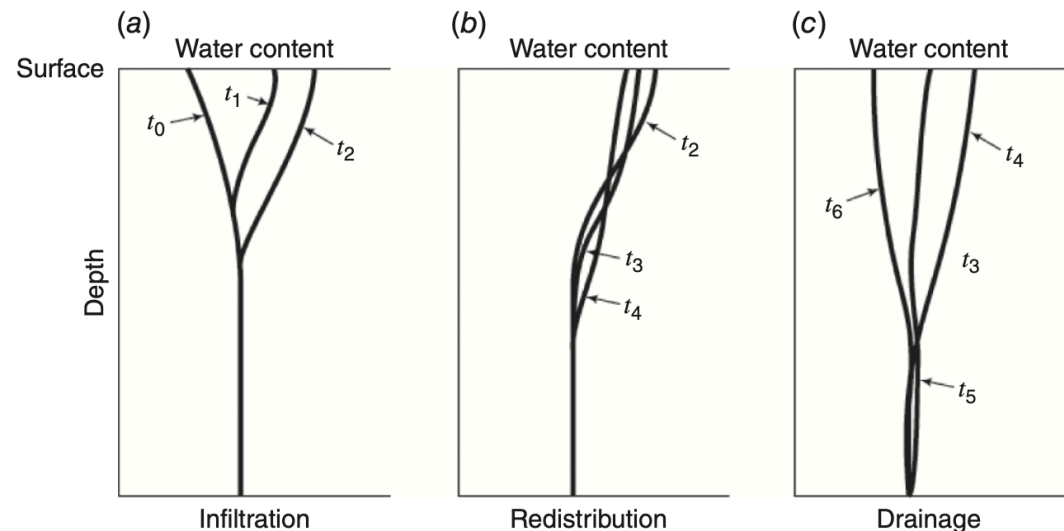


Figure 5: Processes of infiltration redistribution and recharge

Characteristic Curves in the Unsaturated Zone

- In unsaturated media, flow is complex due to **two-phase flow** (air + water).
- Both **volumetric moisture content** (θ) and **hydraulic conductivity** (K) depend on **water content**.
- The **pressure head/moisture content relationship** describes how a sample behaves as water is added or removed.
- As soil dries, void spaces fill with air $\rightarrow$ harder for water to flow.

Key insight

Hydraulic **constants** become hydraulic **functions** in the unsaturated zone.

Water Retention ($\theta(\psi)$) Curves

Water Retention Curve: Relationship between negative pressure head and volumetric water content.

- As θ decreases $\rightarrow \psi$ becomes more negative $\rightarrow$ capillary pressure increases.
- θ is always on **arithmetic scale**, while pressure could be on **log scale**.
- Curves are **nonlinear** regardless of scaling.
- At extremes: small variation in θ results in **big variation** in ψ .
- Minimum θ approaches **residual volumetric water content** θ_r .

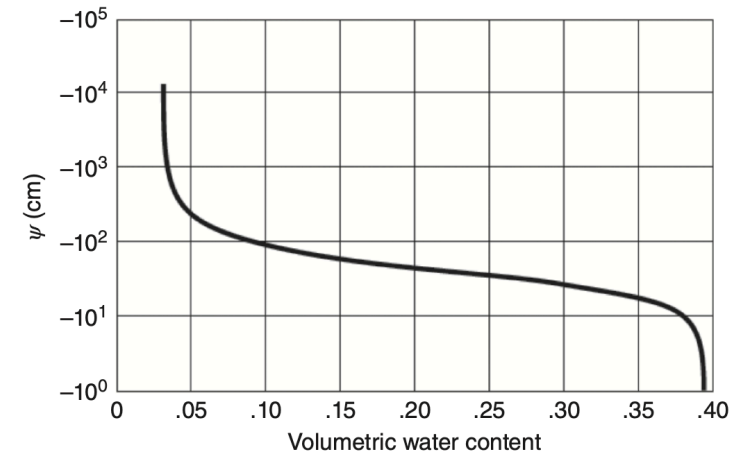


Figure 6: Water retention curve

Effective Saturation

To develop **mathematical relationships**, researchers measure different soil behaviors in the lab and fit equations to the data.

Effective saturation normalizes water content to be independent of soil type:

$$s_e = \frac{\theta - \theta_r}{n - \theta_r}$$

where:

- s_e = effective saturation (0 to 1)
- θ_r = residual volumetric water content
- n = porosity (maximum θ can be)

Two commonly used models:

1. Brooks-Corey relationship
2. van Genuchten equations

Both are written in terms of **effective saturation** s_e .

Brooks-Corey Water Retention Model

Brooks-Corey equation defines effective saturation as:

$$s_e = \begin{cases} \left(\frac{\psi_b}{\psi} \right)^\lambda, & \text{if } \psi < \psi_b \\ 1, & \text{if } \psi \geq \psi_b \end{cases}$$

Parameters:

- ψ_b = **bubbling or air-entry pressure head** [L]
 - Pressure head at which air bubbles start entering the medium
 - Pressure needed to desaturate the **largest pores** in the medium
- λ = **pore-size distribution index** (dimensionless)

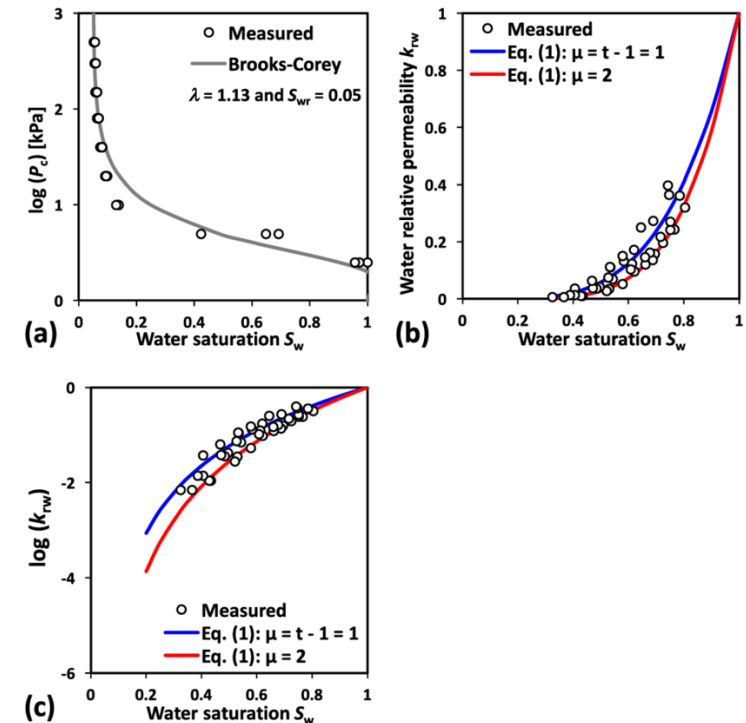


Figure 7: Ghanbarian et al 2017

van Genuchten Water Retention Model

van Genuchten equation is expressed as:

$$s_e = \frac{1}{[1 + (\alpha|\psi|)^\beta]^\gamma}$$

Parameters:

- α = scaling parameter [1/L]
- β = curve shape exponent (dimensionless)
- $\gamma = 1 - \frac{1}{\beta}$ (constraint for van Genuchten)

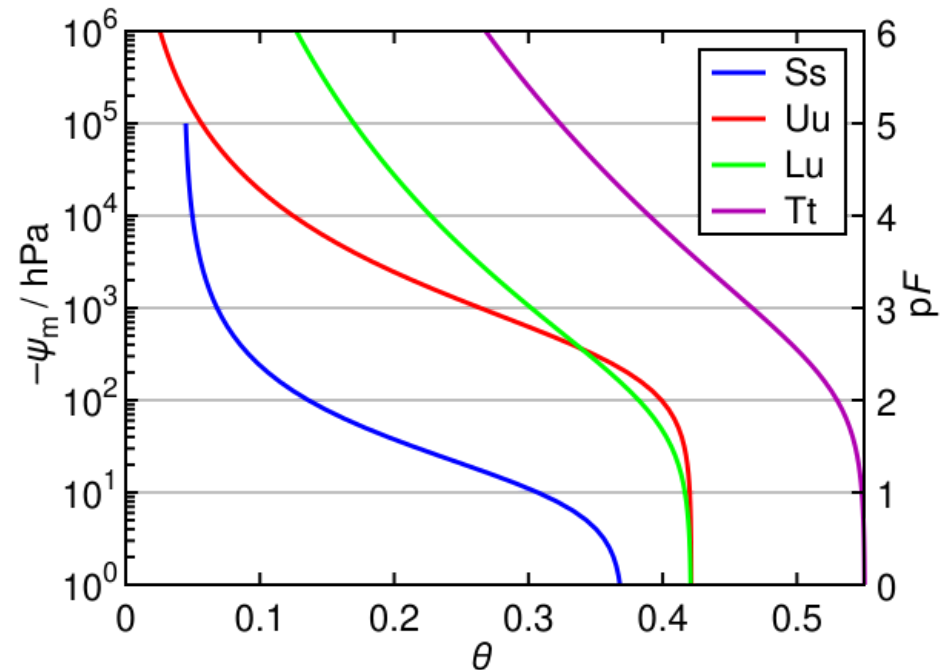


Figure 8: van Genuchten pressure water content curve

Hysteresis and Soil Type Effects

Hysteresis: The actual behavior depends on whether air is entering (drying phase) or water is entering (wetting phase).

- **Drying** → air enters pores, displaces water
- **Wetting** → water enters, displaces air
- Shape is **slightly different** for each process

Soil type impact depends on **pore-size distribution**:

- **Sand** → uniform, large pores → sharp curves
- **Fine sand** → intermediate behavior
- **Silt loam** → broad pore size distribution → gradual curves, higher θ at same ψ

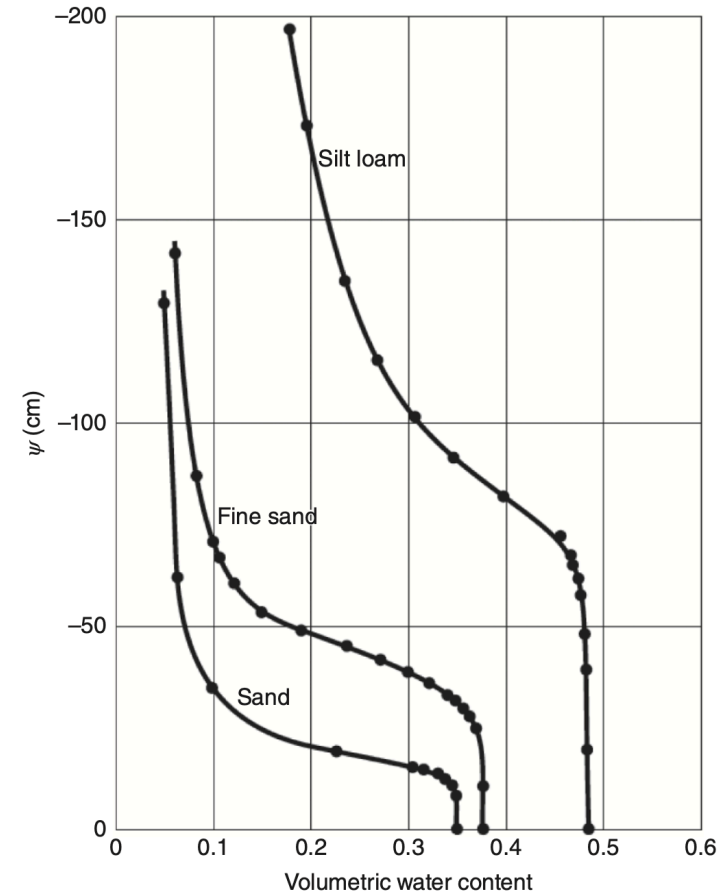


Figure 9: Sand, fine sand, silt loam retention curves

[Back to water-course](#)

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$K(\psi)$ Curves – Unsaturated Hydraulic Conductivity

As expected, for unsaturated media K is **strongly dependent** on the degree of saturation.

- As water content approaches zero $\rightarrow$ negative hydraulic pressure increases more and more.
- Close to residual volumetric content $\theta_r \rightarrow$ hydraulic conductivity becomes **practically zero**.
- Near saturation ($\psi \approx 0$) $\rightarrow K \approx$ maximum value.

Key insight

As soil dries, void spaces fill with air, making it harder for water to flow.

Relative Hydraulic Conductivity

To capture this behavior, we introduce the concept of **relative hydraulic conductivity** or **permeability**.

This shows us the ratio with respect to the saturation property:

$$K_r(\psi) = \frac{K(\psi)}{K_s}$$

where:

- K_r = relative hydraulic conductivity (0 to 1)
- $K(\psi)$ = unsaturated hydraulic conductivity
- K_s = saturated hydraulic conductivity

Purpose:

Shows the fraction of saturated conductivity available at any given pressure head.

Brooks–Corey Model for $K(\psi)$

Going with the same line of thought, the hydraulic conductivity can be written in terms of the negative pressure head:

$$K_r(\psi) = \begin{cases} \left(\frac{\psi_b}{\psi}\right)^{2+3\lambda}, & \psi < \psi_b \\ 1, & \psi \geq \psi_b \end{cases}$$

Parameters:

- ψ_b = bubbling (air-entry) pressure head
- λ = pore-size distribution index

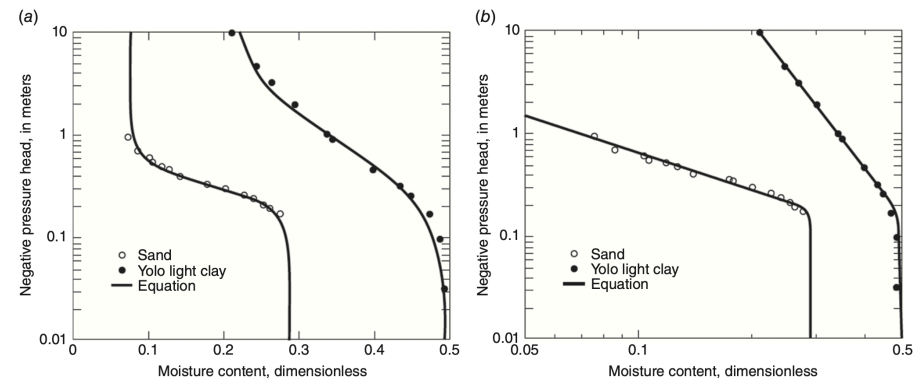


Figure 10: $K(\psi)$ model curves

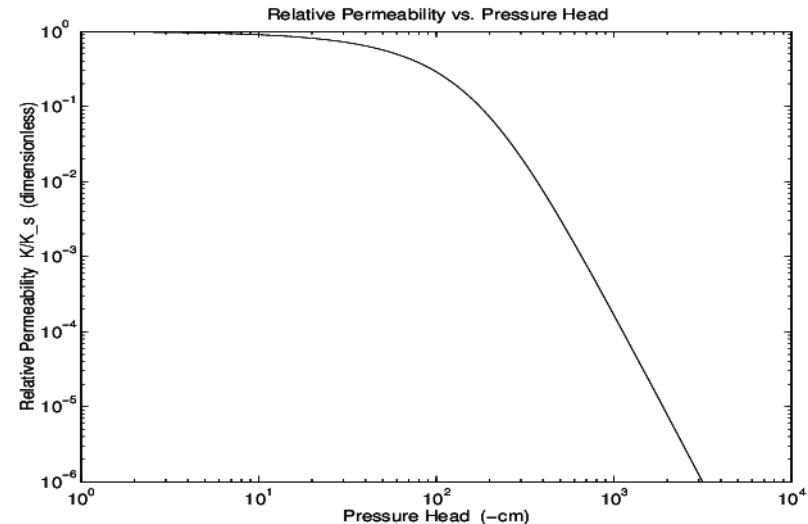
van Genuchten Model for $K(\psi)$

In the same way, we can describe the **van Genuchten relationship** as:

$$K_r(\psi) = \frac{\left[1 - (\alpha|\psi|)^{\beta-1} (1 + (\alpha|\psi|)^{\beta})^{-\gamma}\right]^2}{[1 + (\alpha|\psi|)^{\beta}]^{\gamma/2}}$$

Parameters:

- α = scaling parameter [1/L]
- β = curve steepness parameter
- $\gamma = 1 - \frac{1}{\beta}$ (van Genuchten constraint)



Alternative form in terms of s_e :

$$K_r(s_e) = s_e^l \left[1 - \left(1 - s_e^{1/\beta}\right)^\beta\right]^2$$

with $l \approx 0.5$ (pore connectivity parameter).

Moisture Capacity $C(\psi)$ Curves

The storage properties of an unsaturated soil can be represented by a parameter called **specific moisture capacity** c_m .

Moisture, capillary or differential water capacity is the slope of the water retention curve.

$$c_m = \frac{d\theta}{d\psi}$$

This is basically the **derivative of water content with respect to pressure**.

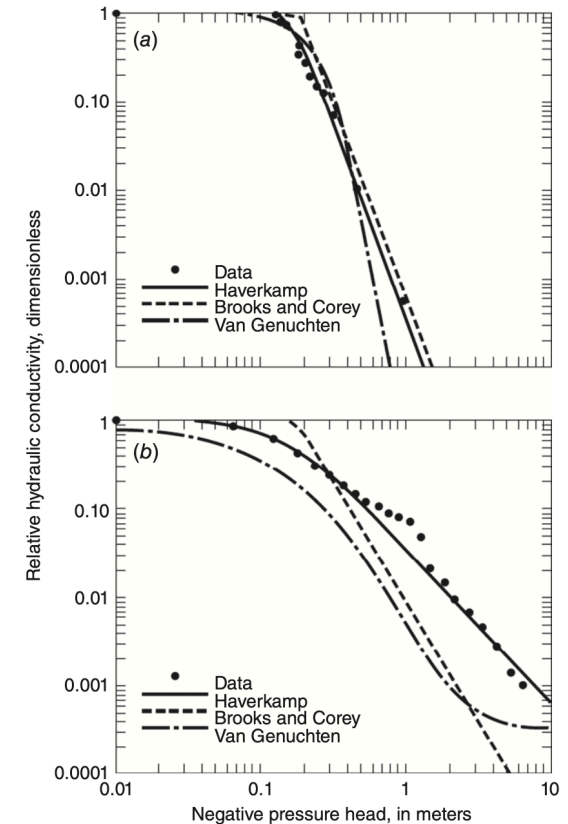


Figure 11: Measured vs modeled $C(\psi)$

Physical meaning: Shows how much θ changes for a given change in ψ .

Models for Moisture Capacity

Brooks-Corey model:

$$c_m(\psi) = \begin{cases} -(n - \theta_r) \frac{\lambda}{\psi_b} \left(\frac{\psi}{\psi_b} \right)^{-(\lambda+1)}, & \psi \leq \psi_b \\ 0, & \psi > \psi_b \end{cases}$$

van Genuchten model:

$$c_m(\psi) = \begin{cases} \frac{\alpha \gamma \beta (n - \theta_r) (\alpha |\psi|)^{\beta-1}}{[1 + (\alpha |\psi|)^{\beta}]^{\gamma+1}}, & \psi < 0 \\ 0, & \psi > 0 \end{cases}$$

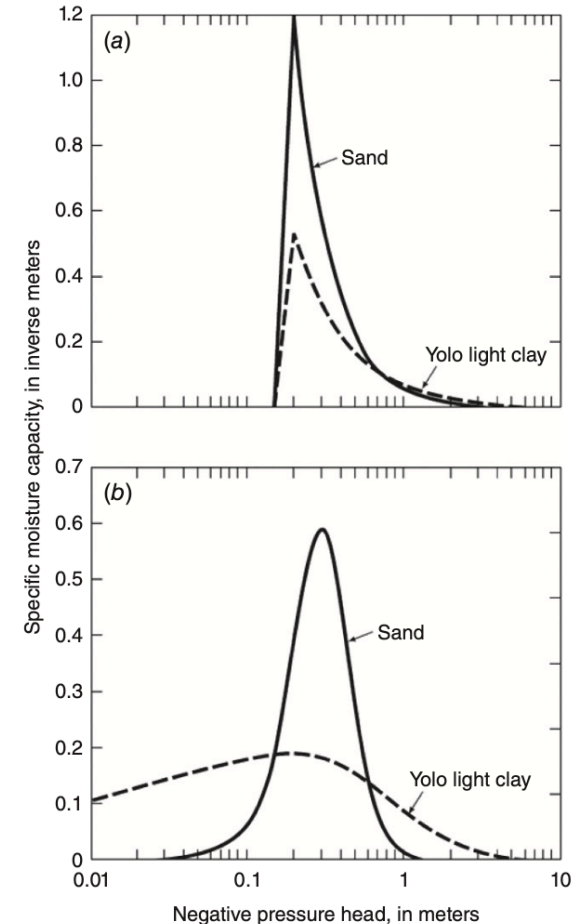


Figure 12: Sand vs clay, Brooks-Corey & van Genuchten

[Back to water-course](#)

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Summary: Characteristic Curves

Three fundamental relationships describe unsaturated flow:

1. **Water Retention Curve** $\theta(\psi)$: How soil holds/releases water as pressure changes
2. **Hydraulic Conductivity Curve** $K(\psi)$: How flow capacity varies with water content
3. **Moisture Capacity Curve** $C(\psi)$: How storage properties change with pressure

Key models:

- **Brooks-Corey**: Uses bubbling pressure ψ_b and pore-size distribution λ
- **van Genuchten**: Uses scaling parameter α , shape parameter β , and constraint $\gamma = 1 - 1/\beta$

All relationships are strongly nonlinear and depend on soil type and pore-size distribution.

Flow Equation in the Unsaturated Zone

Darcy's law (unsaturated form):

$$q = -K(\psi)\nabla h$$

- $K(\psi)$: unsaturated conductivity
- h : hydraulic head

Relative conductivity form: $K(\psi) = K_r(\psi)K_s$

- K_r : relative conductivity (0–1)
- K_s : saturated conductivity

Conservation in the Unsaturated Zone

Combining Darcy + conservation of mass:

$$\frac{\partial}{\partial x_i} \left(K_r(\psi) \left(K_{ij} \frac{\partial h}{\partial x} + K_{iz} \right) \right) = \frac{\partial \theta}{\partial t}$$

- General tensor form with K_{ij} .
- Inflow – outflow = storage change.
- Equivalent to **Richards' equation** for unsaturated flow.

Modified Richards' Equation

The equation can be rewritten with **moisture capacity** and **specific storage**:

$$\frac{\partial}{\partial x} \left(K_r(\psi) K_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_r(\psi) K_y \frac{\partial h}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_r(\psi) K_z \frac{\partial h}{\partial z} + K_r(\psi) K_z \right) = (c_m + sS_s) \frac{\partial h}{\partial t}$$

- c_m : moisture capacity
- s : saturation
- S_s : specific storage

One-Dimensional Case

For vertical flow (z positive downward):

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(K_r(\psi) K_z \left(\frac{\partial \psi}{\partial z} + 1 \right) \right)$$

- Includes both **pressure gradient** and **gravity term** (+1).
- Common in infiltration models.

One-Dimensional Case

For vertical flow (z positive downward):

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left(K_r(\psi) K_z \left(\frac{\partial \psi}{\partial z} + 1 \right) \right)$$

- Includes both **pressure gradient** and **gravity term** (+1).
- Common in infiltration models.

Check-in: How is this similar/different to saturated flow equation?

Velocity in Unsaturated Flow

Linear groundwater (pore water) velocity:

$$v = \frac{q}{\theta} = \frac{q}{sn}$$

- q : Darcy velocity
- θ : volumetric water content
- s : saturation, n : porosity

Practice check: Why does dividing by θ increase velocity compared to q ?